Introduction to Data Science

Group assignment 1

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1 Exercise 1

To show that $A^c \perp\!\!\!\perp B^c$, we wish to show that:

$$P(A^c \cap B^c) = P(A^c) \cdot P(B^c)$$

We begin with the L.H.S., using the formula for the complement of a union to expand the expression $P(A^c \cap B^c)$ as follows:

$$\begin{aligned} P(A^c \cap B^c) &= P((A \cup B)^c) && (1) \\ &= 1 - P(A \cup B) && (2) \\ &= 1 - [P(A) + P(B) - P(A \cap B)] && (3) \end{aligned}$$

Since $A \perp\!\!\!\perp B$, we have that:

$$P(A \cap B) = P(A) \cdot P(B)$$

Thus, the L.H.S. simplifies to:

$$\begin{aligned} P(A^c \cap B^c) &= 1 - [P(A) + P(B) - P(A \cap B)] && (4) \\ &= 1 - P(A) - P(B) + P(A) \cdot P(B) && (5) \end{aligned}$$

We now use the formula for complements to expand the R.H.S. as:

$$\begin{aligned} P(A^c) \cdot P(B^c) &= (1 - P(A)) \cdot (1 - P(B)) && (6) \\ &= 1 - P(A) - P(B) + P(A) \cdot P(B) && (7) \end{aligned}$$

As the L.H.S. = R.H.S., we have shown that:

$$P(A^c \cap B^c) = P(A^c) \cdot P(B^c)$$

and thus, that $A^c \perp\!\!\!\perp B^c$.

2 Exercise 2

Part (a) Solution:
Using the conditional probability formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Let X = number of children with brown hair. We want:

$$P(X \geq 2 | X \geq 1) = \frac{P(X \geq 2 \cap X \geq 1)}{P(X \geq 1)} = \frac{P(X \geq 2)}{P(X \geq 1)}$$

Calculate $P(X \geq 2)$:

$$P(X \geq 2) = P(X = 2) + P(X = 3) \tag{8}$$
$$= \binom{3}{2}\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^1 + \binom{3}{3}\left(\frac{1}{4}\right)^3 \tag{9}$$
$$= 3 \cdot \frac{1}{16} \cdot \frac{3}{4} + 1 \cdot \frac{1}{64} \tag{10}$$
$$= \frac{9}{64} + \frac{1}{64} = \frac{10}{64} \tag{11}$$

Calculate $P(X \geq 1)$:

$$P(X \geq 1) = 1 - P(X = 0) \tag{12}$$
$$= 1 - \left(\frac{3}{4}\right)^3 \tag{13}$$
$$= 1 - \frac{27}{64} = \frac{37}{64} \tag{14}$$

Therefore: $P(X \geq 2 | X \geq 1) = \frac{\frac{10}{64}}{\frac{37}{64}} = \frac{10}{37}$

Part (b) Solution:

Given that the oldest child has brown hair, we need at least one more child (from the two younger children) to have brown hair for a total of at least 2.

Let Y = number of younger children (out of 2) with brown hair.

We want: $P(\text{at least 2 total have brown hair}|\text{oldest has brown hair})$

This is equivalent to: $P(Y \geq 1)$ where $Y \sim \text{Binomial}(2, \frac{1}{4})$

$$P(Y \geq 1) = 1 - P(Y = 0) \tag{15}$$
$$= 1 - \binom{2}{0}\left(\frac{1}{4}\right)^0\left(\frac{3}{4}\right)^2 \tag{16}$$
$$= 1 - 1 \cdot 1 \cdot \frac{9}{16} \tag{17}$$
$$= 1 - \frac{9}{16} = \frac{7}{16} \tag{18}$$

Therefore: $P(\text{at least 2 have brown hair}|\text{oldest has brown hair}) = \frac{7}{16}$

3 Exercise 3

$R = \sqrt{X^2 + Y^2}$. For $0 \leq r \leq 1$, the cumulative distribution function (CDF) is

$$F_R(r) = P(R \leq r) = \frac{\text{Area of disk of radius } r}{\text{Area of the unit disk}} = \frac{r^2\pi}{1^2\pi} = r^2.$$

Therefore,

$$F_R(r) = \begin{cases} 0, & r < 0, \\ r^2, & 0 \leq r \leq 1, \\ 1, & r > 1. \end{cases}$$

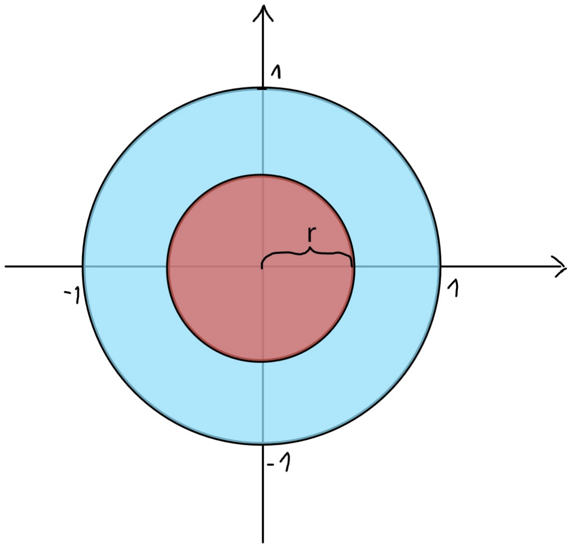

Figure 1: Image depicting the idea of solving exercise 3

Differentiating gives the probability density function PDF:

$$f_R(r) = \frac{d}{dr} F_R(r) = \begin{cases} 2r, & 0 \le r \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

4 Exercise 4

Problem

A fair coin is tossed until a head appears. Let X be the number of tosses required. What is the expected value of X?

Solution

The experiment consists of independent tosses of a fair coin. Let a "success" be the appearance of a head. Since the coin is fair, the probability of success in each toss is

$$p = \frac{1}{2}.$$

Since the random variable X belongs to the Bernoulli distribution, the number of tosses until the first head, has the probability mass function

$$\mathbb{P}(X = k) = (1-p)^{k-1} p = \left(\tfrac{1}{2}\right)^k, \qquad k = 1, 2, 3, \ldots$$

(Here we used $1 - p = \frac{1}{2}$.)
The expectation of X is

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k \, \mathbb{P}(X = k) = \sum_{k=1}^{\infty} k \left(\tfrac{1}{2}\right)^k$$

because it's just the sum of the number of tosses required (k) multiplied by the probability of this many tosses required (which is just the pmf).
To evaluate this infinite series, we will use the property of infinite series.

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}, \qquad |r| < 1.$$

Differentiate both sides with respect to r:

$$\sum_{k=1}^{\infty} k r^{k-1} = \frac{1}{(1-r)^2}.$$

Multiply by r to obtain

$$\sum_{k=1}^{\infty} k r^k = \frac{r}{(1-r)^2}, \qquad |r| < 1.$$

Set $r = \frac{1}{2}$ to get

$$\sum_{k=1}^{\infty} k \left(\tfrac{1}{2}\right)^k = \frac{\frac{1}{2}}{\left(1-\frac{1}{2}\right)^2} = \frac{\frac{1}{2}}{\left(\frac{1}{2}\right)^2} = 2.$$

Therefore

$$\mathbb{E}[X] = 2.$$

5 Exercise 5

a) Proof

Task. Let $X_1, \ldots, X_n$ be i.i.d. from Bernoulli(p). (a) Let $\alpha > 0$ be fixed and define

$$\varepsilon_n = \sqrt{\frac{1}{2n} \log \frac{2}{\alpha}}.$$

Let

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^{n} X_i, \quad I_n = [\hat{p}_n - \varepsilon_n,\ \hat{p}_n + \varepsilon_n].$$

Use Hoeffding's inequality to show that

$$\Pr(p \in I_n) \geq 1 - \alpha.$$

Proof. Let $X_1, \ldots, X_n$ be independent random variables with $X_i \in [0,1]$, and let $\hat{p}_n = \overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.
By Hoeffding's inequality for bounded independent random variables (two-sided form),

$$\Pr\Big(\big|\overline{X}_n - \mathbb{E}[\overline{X}_n]\big| \geq \varepsilon\Big) \leq 2\exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right),$$

where each $X_i \in [a,b]$. In the Bernoulli case $a = 0$, $b = 1$, so $(b-a)^2 = 1$ and hence

$$\Pr\left(|\overline{X}_n - \mathbb{E}[\overline{X}_n]| \geq \varepsilon\right) \leq 2\exp(-2n\varepsilon^2). \tag{19}$$

For Bernoulli(p) variables we have $\mathbb{E}[\overline{X}_n] = p$, so (19) becomes

$$\Pr\Big(|\hat{p}_n - p| \geq \varepsilon\Big) \leq 2\exp(-2n\varepsilon^2).$$

Now substitute $\varepsilon = \varepsilon_n$. Then, we can modify the right-hand side

$$2\exp\Big(-2n\varepsilon_n^2\Big) = 2\exp\Big(-2n \cdot \frac{1}{2n}\log\frac{2}{\alpha}\Big) = 2\exp\Big(-\log\frac{2}{\alpha}\Big) = \alpha,$$

so

$$\Pr\Big(|\hat{p}_n - p| \geq \varepsilon_n\Big) \leq \alpha.$$

Taking the complement of the set gives

$$\Pr\Big(|\hat{p}_n - p| < \varepsilon_n\Big) \geq 1 - \alpha.$$

Since $|\hat{p}_n - p| < \varepsilon_n$ is exactly the event $p \in I_n = [\hat{p}_n - \varepsilon_n, \hat{p}_n + \varepsilon_n]$, we conclude

$$\Pr(p \in I_n) \geq 1 - \alpha,$$

which proves the claim.

b) Simulation study

For points b) and c) we have written a Python script and ran it in a Jupyter notebook. The script looks as follows:

```
import numpy as np
import matplotlib.pyplot as plt
alpha = 0.05
p = 0.4

reps = 10000
sample_sizes = [10, 100, 1000, 10000]
rng = np.random.default_rng(23)
coverages = []

lengths = []

for n in sample_sizes:
    draws = rng.binomial(1, p, size=(reps,n))

    phats = draws.mean(axis=1)
    eps = np.sqrt((1 / (2 * n)) * np.log(2 / alpha))

    # Construct confidence intervals
    lower = phats - eps
    upper = phats + eps

    length = upper - lower
    lengths.append(length.mean())
```

```
covered = (p >= lower) & (p <= upper)
coverage = covered.mean()

coverages.append(coverage)

print(f"n={n}, epsilon={eps:.4f},
        coverage={coverage:.4f}")
```

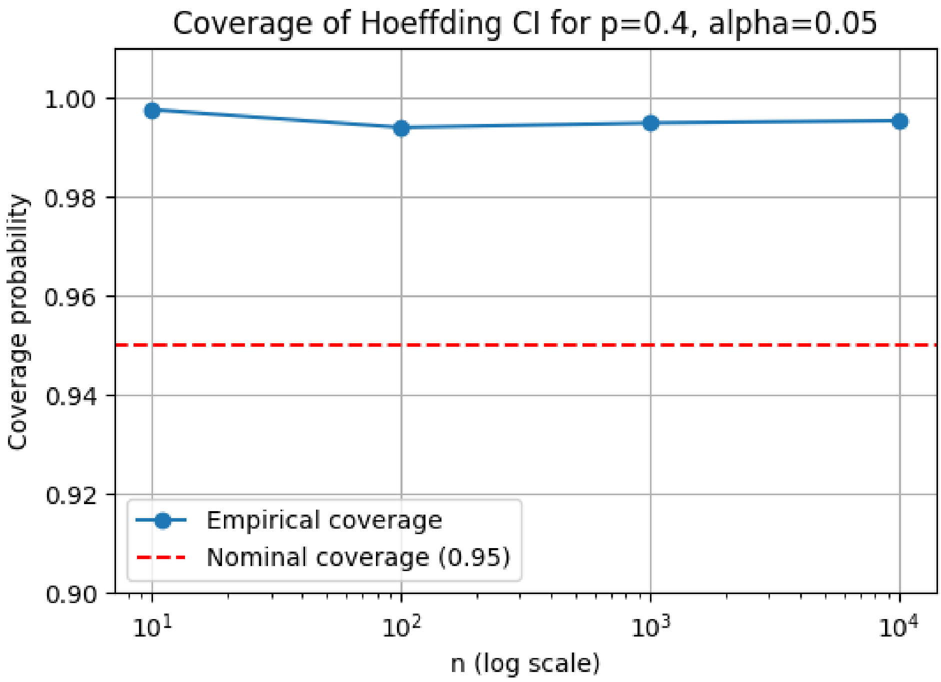

Figure 2: Coverage of Hoeffding confidence interval

The plots depicting results of the simulation are visible in Figure 2 (with log scale on x-axis for better readability). It demonstrates that the empirical coverage of the Hoeffding confidence interval consistently stays above the nominal coverage level of $1 - \alpha = 0.95$, for all sample sizes. We can confirm that the Hoeffding bound is a conservative one. It provides a robust guarantee on the coverage, but may produce wider than necessary intervals.

c) Length of confidence interval

The average interval length is visible in Figure 3 (with both axes presented in log scale. There is a negative relationship between the length and the sample size. We are more confident about the results we obtain as we get more samples. This demonstrates the fundamental trade-off in statistical inference: larger sample sizes lead to more precise estimates (narrower intervals) at a higher cost.

d) Disease simulation For point d), the script is similar; however, we check if p_new falls into the confidence interval dictated by p_old.

```
import numpy as np
import matplotlib.pyplot as plt
```

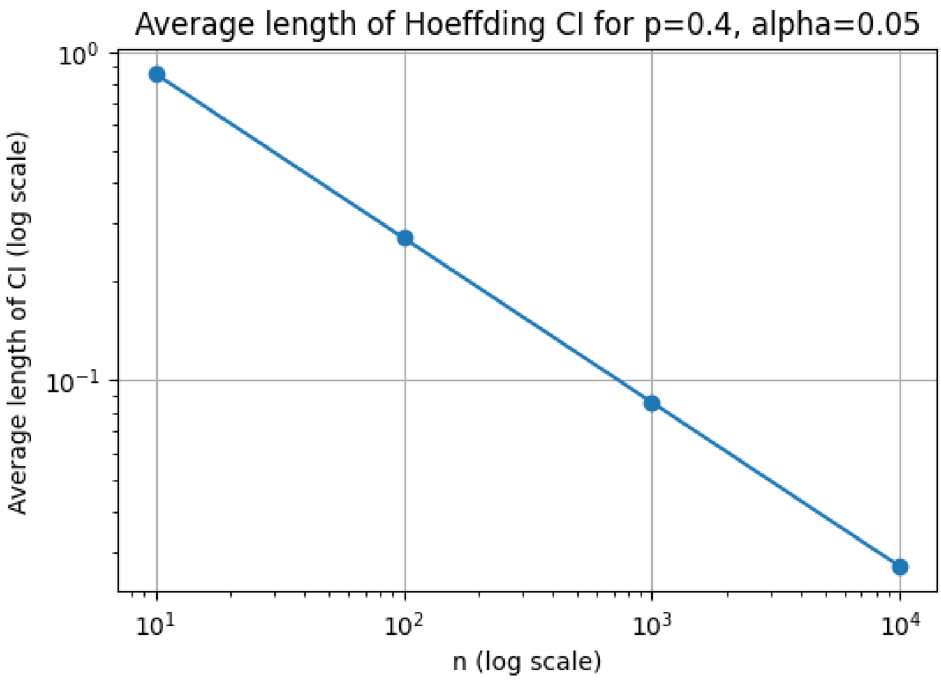

Figure 3: Interval length against number of samples

```
alpha = 0.05
p_old = 0.4  # The true proportion
p_new = 0.5  # The new true proportion

reps = 10000
sample_sizes = [10, 100, 1000, 10000]
rng = np.random.default_rng(23)

correct_decision_probs = []

for n in sample_sizes:
    # Draw samples from the original distribution (p_old = 0.4)
    draws = rng.binomial(1, p_old, size=(reps, n))

    # Calculate sample means
    phats = draws.mean(axis=1)

    eps = np.sqrt((1 / (2 * n)) * np.log(2 / alpha))

    lower = phats - eps
    upper = phats + eps


    correct_decisions = (p_new >= lower) & (p_new <= upper)

    correct_prob = correct_decisions.mean()
    correct_decision_probs.append(correct_prob)
```

```
print(f"n={n}, Probability of correct decision
    {correct_prob:.4f}")
```

The results are presented in Figure 4. It is visible that as n increases, the probability of a correct decision decreases. As we have more samples the confidence interval is shrinking and its centre is close to 0.4 so it becomes less likely to contain the new true proportion.

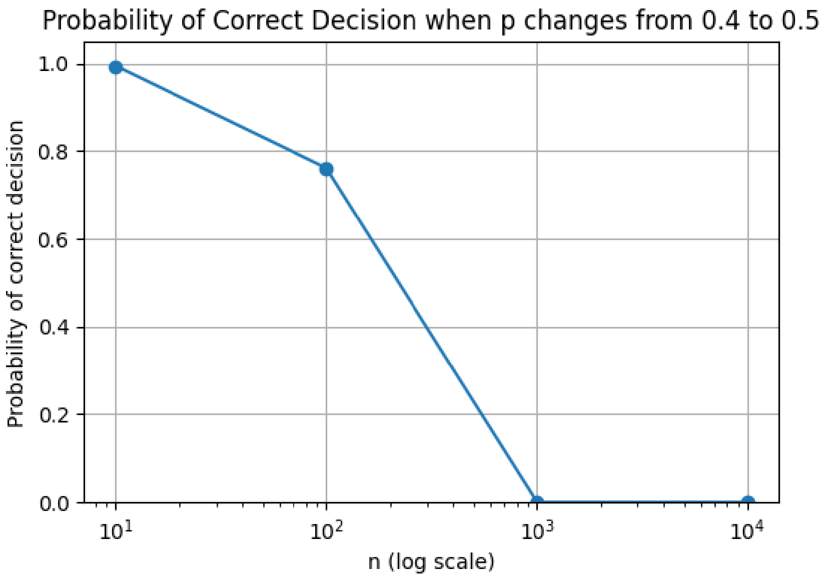

Figure 4: Probability that the new p falls into the confidence region dictated by the old p.